作业 5 期末大作业

第三版第 14 章问题：14.1, 14.9

或　第四版第 14 章问题：14.1, 14.10

HFSS Simulation

Use HFSS to simulate the input impedance of the antenna in Prob. 14.9 in version III of the textbook or Prob. 14.10 in version IV of the textbook, where

- The patch antenna is feed by a 50Ohm microstrip line(use Equ14.19 to calculate the width of the feed line)
- The ground plane should be larger than the patch(can be 5L*5W)
- Both patch and ground are PEC
- Lumped port is used for the excitation
- Frequency range: 8GHz-14GHz

Determine:

- The difference between HFSS's resonant frequency and 10GHz, why is there difference?
- Calculate the E-and H-plane radiation at the HFSS resonant frequency